Question 3 One Step of Gradient Descent in a Deep Neural Network

Given: Input $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{R}^2$, true class $y = 1$, learning rate $\eta = 0.1$

Network weights:

$$W^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad W^{(2)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad W^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(3)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

ReLU derivative convention: $\mathrm{ReLU}'(t) = \begin{cases} 1 & t > 0 \\ 0 & t \le 0 \end{cases}$

Part A: Softmax with Cross-Entropy Loss

(a) Forward Pass

Layer 1 — Affine + ReLU:

$$z^{(1)} = W^{(1)}x + b^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$a^{(1)} = \mathrm{ReLU}\big(z^{(1)}\big) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Layer 2 — Affine + ReLU:

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1-1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$a^{(2)} = \mathrm{ReLU}\big(z^{(2)}\big) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Layer 3 — Output (affine only):

$$s = W^{(3)}a^{(2)} + b^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Softmax probabilities:

$$q_i = \frac{e^{s_i}}{\sum_j e^{s_j}}, \qquad e^{s_1} + e^{s_2} = e^0 + e^2 = 1 + 7.3891 = 8.3891$$

$$\boxed{q_1 = \frac{e^0}{e^0 + e^2} = \frac{1}{8.3891} \approx 0.1192, \qquad q_2 = \frac{e^2}{e^0 + e^2} = \frac{7.3891}{8.3891} \approx 0.8808}$$

(b) Cross-Entropy Loss

$$L_{\mathrm{CE}} = -\log q_y = -\log q_1 = -\log(0.1192) \approx \boxed{2.1269}$$

(c) Gradient w.r.t. Logits s

For softmax + cross-entropy, the gradient simplifies to:

$$\frac{\partial L_{\mathrm{CE}}}{\partial s_i} = q_i - \mathbf{1}[i = y]$$

$$\frac{\partial L_{\mathrm{CE}}}{\partial s} = \begin{bmatrix} q_1 - 1 \\ q_2 - 0 \end{bmatrix} = \begin{bmatrix} 0.1192 - 1 \\ 0.8808 \end{bmatrix} = \boxed{\begin{bmatrix} -0.8808 \\ 0.8808 \end{bmatrix}} \triangleq \delta^s$$

(d) Backpropagation

Layer 3 gradients:

$$\nabla_{W^{(3)}} = \delta^s \cdot \left(a^{(2)}\right)^T = \begin{bmatrix} -0.8808 \\ 0.8808 \end{bmatrix} \begin{bmatrix} 0 & 2 \end{bmatrix} = \boxed{\begin{bmatrix} 0 & -1.7616 \\ 0 & 1.7616 \end{bmatrix}}$$

$$\nabla_{b^{(3)}} = \delta^s = \boxed{\begin{bmatrix} -0.8808 \\ 0.8808 \end{bmatrix}}$$

Propagate back through Layer 3:

$$\delta^{a^{(2)}} = \left(W^{(3)}\right)^T \delta^s = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -0.8808 \\ 0.8808 \end{bmatrix} = \begin{bmatrix} -0.8808 \\ 0.8808 \end{bmatrix}$$

Apply ReLU$'$ at $z^{(2)} = [0,\ 2]^T$:

$$\mathrm{ReLU}'\left(z^{(2)}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \implies \delta^{z^{(2)}} = \delta^{a^{(2)}} \odot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.8808 \\ 0.8808 \end{bmatrix} \odot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.8808 \end{bmatrix}$$

Layer 2 gradients:

$$\nabla_{W^{(2)}} = \delta^{z^{(2)}} \cdot \left(a^{(1)}\right)^T = \begin{bmatrix} 0 \\ 0.8808 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 0 & 0 \\ 0.8808 & 0.8808 \end{bmatrix}}$$

$$\nabla_{b^{(2)}} = \delta^{z^{(2)}} = \boxed{\begin{bmatrix} 0 \\ 0.8808 \end{bmatrix}}$$

Propagate back through Layer 2:

$$\delta^{a^{(1)}} = \left(W^{(2)}\right)^T \delta^{z^{(2)}} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0.8808 \end{bmatrix} = \begin{bmatrix} 0.8808 \\ 0.8808 \end{bmatrix}$$

Apply ReLU$'$ at $z^{(1)} = [1,\ 1]^T$:

$$\delta^{z^{(1)}} = \begin{bmatrix} 0.8808 \\ 0.8808 \end{bmatrix}$$

Layer 1 gradients:

$$\nabla_{W^{(1)}} = \delta^{z^{(1)}} \cdot x^T = \begin{bmatrix} 0.8808 \\ 0.8808 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 0.8808 & 0.8808 \\ 0.8808 & 0.8808 \end{bmatrix}}$$

$$\nabla_{b^{(1)}} = \delta^{z^{(1)}} = \boxed{\begin{bmatrix} 0.8808 \\ 0.8808 \end{bmatrix}}$$

(e) Gradient Descent Update

$$W^{(3)}_{\text{new}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0.1 \begin{bmatrix} 0 & -1.7616 \\ 0 & 1.7616 \end{bmatrix} = \boxed{\begin{bmatrix} 1 & 0.1762 \\ 0 & 0.8238 \end{bmatrix}}$$

$$b^{(3)}_{\text{new}} = \boxed{\begin{bmatrix} 0.0881 \\ -0.0881 \end{bmatrix}}$$

$$W^{(2)}_{\text{new}} = \boxed{\begin{bmatrix} 1 & -1 \\ 0.9119 & 0.9119 \end{bmatrix}}$$

$$b^{(2)}_{\text{new}} = \boxed{\begin{bmatrix} 0 \\ -0.0881 \end{bmatrix}}$$

$$W^{(1)}_{\text{new}} = \boxed{\begin{bmatrix} 0.9119 & -0.0881 \\ -0.0881 & 0.9119 \end{bmatrix}}$$

$$b^{(1)}_{\text{new}} = \boxed{\begin{bmatrix} -0.0881 \\ -0.0881 \end{bmatrix}}$$

Part B: Multiclass Hinge Loss

(Same forward pass: $s = [0,\ 2]^T$, true class $y = 1$.)

(a) Hinge Loss

$$L_{\text{hinge}} = \sum_{j \neq y} \max(0,\ s_j - s_y + 1)$$

$$L_{\text{hinge}} = \boxed{3}$$

(b) Gradient w.r.t. Logits s

$$\frac{\partial L_{\text{hinge}}}{\partial s} = \boxed{\begin{bmatrix} -1 \\ +1 \end{bmatrix}}$$

(c) Backpropagation

$$\nabla_{W^{(3)}} = \boxed{\begin{bmatrix} 0 & -2 \\ 0 & 2 \end{bmatrix}}$$

$$\nabla_{b^{(3)}} = \boxed{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}$$

$$\nabla_{W^{(2)}} = \boxed{\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}}$$

$$\nabla_{b^{(2)}} = \boxed{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}$$

$$\nabla_{W^{(1)}} = \boxed{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}$$

$$\nabla_{b^{(1)}} = \boxed{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

(d) Gradient Descent Update

$$W^{(3)}_{\text{new}} = \boxed{\begin{bmatrix} 1 & 0.2 \\ 0 & 0.8 \end{bmatrix}}$$

$$b^{(3)}_{\text{new}} = \boxed{\begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}}$$

$$W^{(2)}_{\text{new}} = \boxed{\begin{bmatrix} 1 & -1 \\ 0.9 & 0.9 \end{bmatrix}}$$

$$b^{(2)}_{\text{new}} = \boxed{\begin{bmatrix} 0 \\ -0.1 \end{bmatrix}}$$

$$W^{(1)}_{\text{new}} = \boxed{\begin{bmatrix} 0.9 & -0.1 \\ -0.1 & 0.9 \end{bmatrix}}$$

$$b^{(1)}_{\text{new}} = \boxed{\begin{bmatrix} -0.1 \\ -0.1 \end{bmatrix}}$$